Classification by Decision Trees - RapidMiner

Introduction

The goal of this activity is to perform classification on Traffic dataset using Decision Trees. The task is to predict whether accident sevirty. Also, find the accuracy of your model.

Download : //www.kaggle.com/datasets/denkuznetz/traffic-accident-prediction

Steps:

1. **Import & Save:** Import the Traffic dataset into RapidMiner and save it.
2. **Filter Missing Values:** Identify your class column and use the **"Filter Examples"** operator to remove any missing values. Then, check your dataset and report the number of instances **before and after** filtering.

Class – Accident

Missing before – 840 after 798

3. **Check for Duplicates:** Identify any duplicate entries. If duplicates exist, specify how many.

786

4. **Check for Missing Values:** Identify the correct data type for each attribute (including your class column) and list the number of missing values for each.
5. **Categorize Data & Handling missing values:** Classify the dataset into quantitative and qualitative attributes in a table and handle any missing values accordingly.
6. **Set Roles & Create Copies:** Assign roles to attributes.
7. **Prepare Data for Decision Tree:** In the **Decision Tree (DT) chain**, convert all attributes to nominal values by creating a new column using the **"Generate Attribute"** operator. Then, **Select** relevant attributes for analysis.
8. **Apply Decision Tree Method:** Use either **10-fold cross-validation** or a **percentage split** for the Decision Tree model.
9. **Tune Decision Tree Parameters:** Read about Decision Tree parameters such as **confidence, minimal leaf size, minimal size for split, and pruning** using the **"i"** button next to each parameter. Experiment with different values, and after a few trials, determine the best settings.
   - In your answer document, include at least **three trials** with their **parameter settings, screenshot of performance vector (**showing the accuracy and confusion matrix**).**